

ERROR IN METHOD OF MOMENTS

Can we estimate the error in our estimates of θ ? $\left(\begin{array}{l} \theta = \lambda \text{ for Poisson} \\ \theta = [\mu, \sigma^2] \text{ for normal} \end{array} \right)$

Ⓐ Sometimes yes we can derive analytic expressions

For $\bar{X} \sim \text{normal}(\mu, \sigma^2)$ we know $\hat{\mu} = \bar{X} \sim \text{normal}(\mu, \sigma^2/\sqrt{n})$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Ⓑ Sometimes we resort to simulations (Monte Carlo)

① Take estimated $\hat{\theta} \rightarrow$ estimated best fit pdf $f(x, \hat{\theta})$

② Run many sims $\{\bar{X}_i\}^k \sim f(x, \hat{\theta})$

$$\{\bar{X}_i\}^k \rightarrow \hat{\theta}_k$$

③ History $\{\hat{\theta}_k\}_{k=1}^{1000}$

BOOTSTRAPPING

Ex Ⓐ for Poisson(λ)

$$E(\hat{\lambda}) = \frac{1}{n} \sum_{i=1}^n E(\bar{X}_i) = \lambda$$

$$\text{Var}(\hat{\lambda}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(\bar{X}_i) = \frac{\lambda}{n}$$

$$\sigma_{\hat{\lambda}} = \sqrt{\lambda/n} \Rightarrow S_{\hat{\lambda}} = \sqrt{\frac{\hat{\lambda}}{n}} \quad \underline{\underline{\text{BOOTSTRAP}}}$$